PolySkill: Learning Generalizable Skills Through Polymorphic Abstraction

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Paper version: arXiv 2510.15863v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Existing methods for skill learning often create skills that are over-specialized to a single website and fail to generalize.*

Code skills tied to one site's layout are hard to reuse on another site.

Motivation: three gaps in skill induction

1. **Over-specialised.** Code hard-wires one site's layout.
2. **Low reuse.** Under 9% (ASI), 3% (SkillWeaver) on unseen sites.
3. **Unstable.** SkillWeaver degrades as self-set tasks narrow.

Evaluation setup

WebArena · 812 | Mind2Web · 2,350 tasks | GPT-4.1 | Claude-3.7-Sonnet | Qwen3-Coder-480B-A35B | GLM-4.5 | Mind2Web: 3 runs | 1009 training tasks

- **WebArena** is execution-checked; **Mind2Web** is scored by a GPT-4.1 judge.
- **Held out:** Mind2Web cross-website and cross-domain splits.

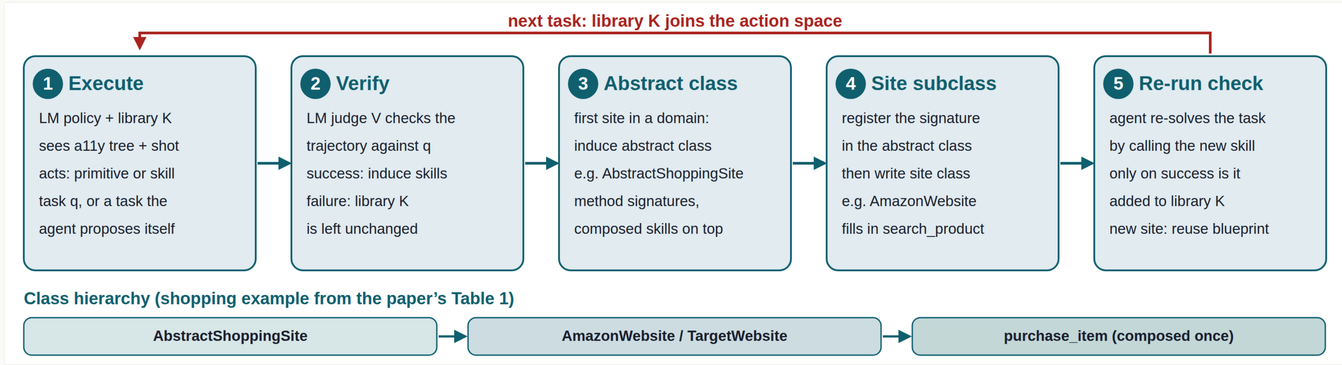
Our read (CUA Radar)

69	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out splits; baselines; cost; judge check; code.
- **Flags:** AP5 (data overlap); variance, budget match not recorded.
- Pattern C19. Authors' numbers, not rerun.

Method: an abstract class per domain, one subclass per site

- **Sees / acts:** a11y tree + viewport screenshot in; BrowserGym primitives or skill calls out.
- **Skill:** abstract domain class (AbstractShoppingSite) plus a subclass per site (AmazonWebsite).
- **Signal:** LM judge verifies the run; the skill is re-run before it is stored. Prompting, no weight updates.



Our redraw (not the paper's figure): the induction loop and the class hierarchy from the paper's shopping example.

$$\max_{\pi_L,\,K} \mathbb{E}_{q\sim Q}\left[\, g(\tau, q) - \gamma\,|\tau| \,\right]$$

Efficiency-aware objective; the paper uses it to guide prompting, not as a loss

Key results: WebArena success rate, 812 tasks (%)

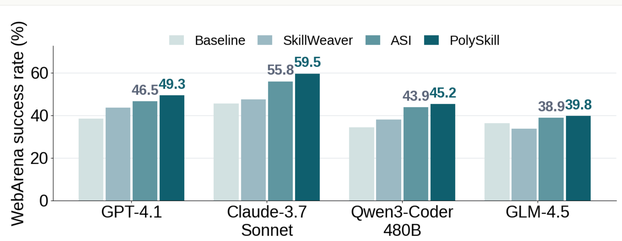


Table 8, redrawn: four backbones; no std. reported.

WebArena avg: 49.3 vs ASI 46.5 (GPT-4.1)

SO WHAT → *Paper's own difference vs ASI: +2.8 / +3.7 / +1.3 / +0.9. Below ASI on some per-site columns.*

Mind2Web cross-task: skill-scope variants (%)

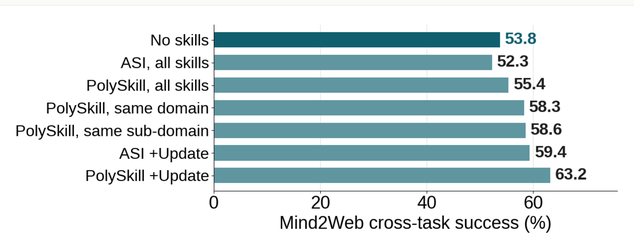


Table 6, redrawn (GPT-4.1): varies skill scope and updating, not the polymorphic split.

SO WHAT → *Same-domain 58.3 and +Update 63.2 are the higher PolySkill rows; ASI +Update is 59.4. Static all-skills 55.4 vs ASI 52.3 and no-skill 53.8. These rows vary skill scope and updating, so they do not isolate the abstract/concrete split.*

Headline Numbers

49.3%
WebArena avg, GPT-4.1
vs ASI 46.5, SkillWeaver 43.6, base 38.5 (Table 8)

- **63.2%** M2W x-task +Update
- **59.5%** WebArena, Claude-3.7
- **20.4%** reuse, WA shopping

The paper's own controls

CONTROL (GPT-4.1)	POLYSKILL	COMPARED WITH
Mind2Web cross-domain, static	60.1	ASI 57.3; base 62.3
Skills: x-task, st. / +Upd	43 / 47	ASI 50 / 66
Action-gen tokens per task	49,710.3	base 55,367.4; ASI 53,840.5
Self-guided, WA shopping	43.1	curric. 42.1; base 37.4

Takeaway

Higher WebArena average than ASI on all four backbones; Mind2Web gains are largest with online +Update (63.2 vs 59.4); static cross-domain stays below no-skill.

Skills 43 vs 50 (static) | +Update skills 47 vs 66 | WebArena: no std. dev. | Mind2Web: 3 runs

Scan to Read

Paper | Code